Practical-01

Understand and implement the different ER diagram notation with their relationship and Cardinalities.

Objective: -To understand different ER diagram notations and the relationships and cardinalities used in database modeling.

An Entity Relationship (ER) diagram is a graphical tool used to design and represent the structure of a database. It helps in visualizing entities, their attributes, and how they are connected with each other. ER diagrams are mainly used during the planning phase of database development.

- **Basic ER Diagram Notations**

i.) **Entity**

- Represented by a rectangle
- An entity is a real-world object or concept
- Example: Student, Employee, Department

ii.) **Attribute**

- Represented by an oval
- Describes properties of an entity
- Example: Name, Age, Salary

iii.) **Primary Key Attribute**

- Underlined oval
- Uniquely identifies each record
- Example: Roll_No, Emp_ID

iv.) **Relationship**

- Represented by a diamond
- Shows connection between entities

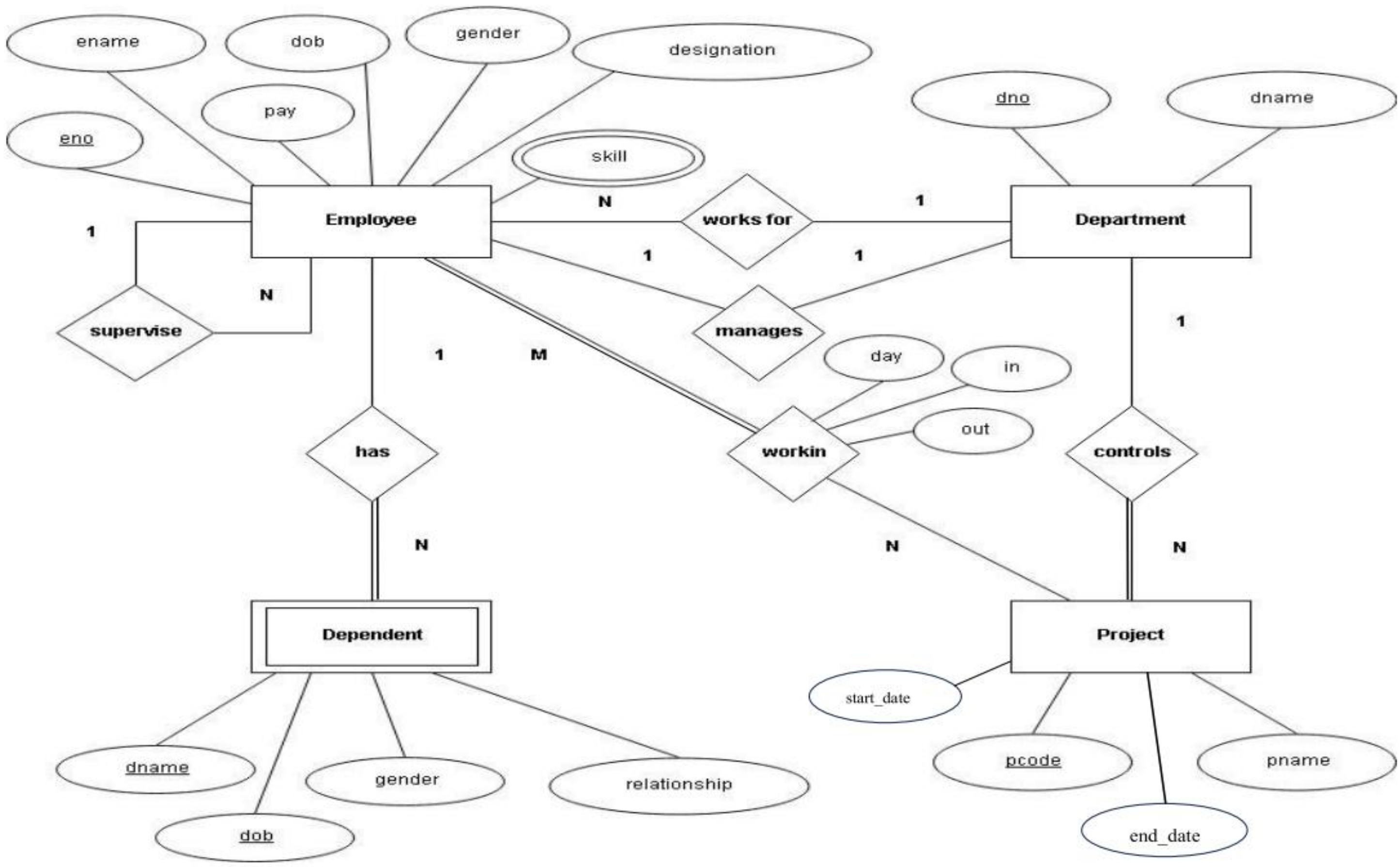
Practical-02

Creating ER Diagram for company Database. Company database have entities like employee, departments, projects and dependents also implement the relationship and cardinalities between the entities with their relevant attribute.

Objective: - Creating ER Diagram for company Database. Company database has entities like employee, departments, projects and dependents also implement the relationship and cardinalities between the entities with their relevant attribute.

The ER diagram identifies the main entities involved in the organization, their attributes, and the relationships among them along with appropriate cardinalities. This model helps in understanding how different components of the company database are connected.

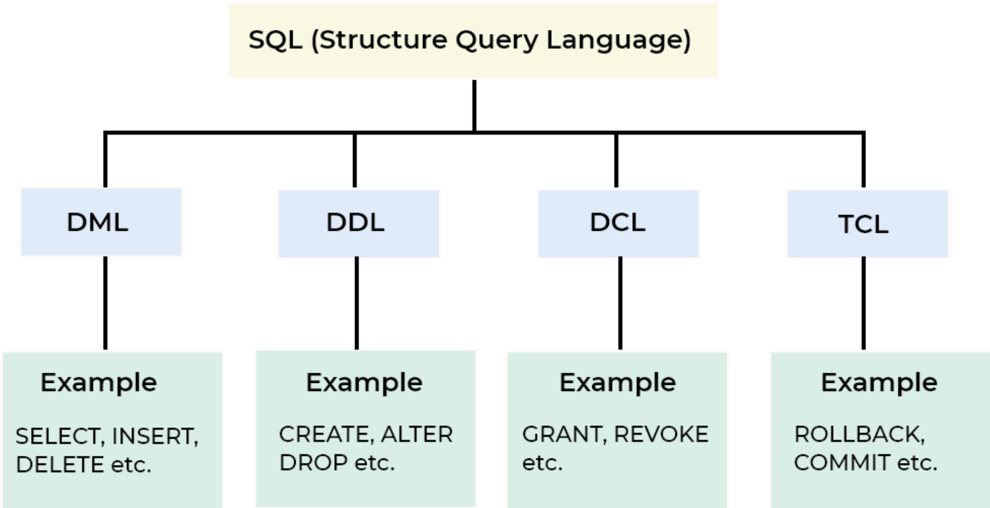
<u>ER Diagram for Company Database:</u>



Practical-03

Implement DDL, DML, DCL & TCL commands.

Objective: - To study and implement different SQL commands including DDL, DML, DCL and TCL.

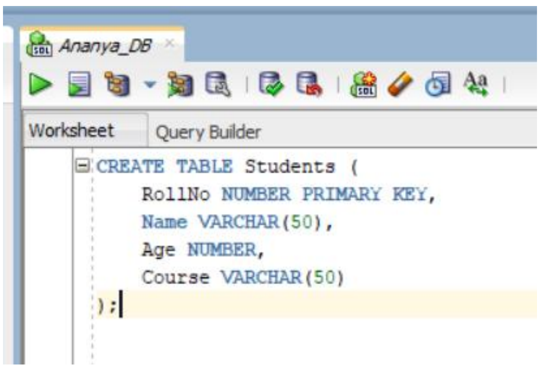


DDL Commands (Data Definition Language): -

- DDL commands are used to create and modify database structure.
- DDL (Data Definition Language) consists of SQL commands that can be used for defining, altering and deleting database structures such as tables, indexes and schemas.

1. **<u>CREATE:</u>** It is used to create objects in the database

Code:




```sql
CREATE TABLE Students (
    RollNo NUMBER PRIMARY KEY,
    Name VARCHAR(50),
    Age NUMBER,
    Course VARCHAR(50)
);
```

Output:

```
Table STUDENTS created.
```

Practical-04

Implementation of I/O Constraint: Primary Key, composite primary key, Foreign Key with on delete set null and on delete set null constraint, Unique Key.

Objective: -To implement integrity constraints in a database including Primary Key, Composite Primary Key, Foreign Key with ON DELETE SET NULL, and Unique Key to maintain data accuracy and consistency.

Primary Key-

- A primary key uniquely identifies each record in a table.
- It does not allow duplicate or NULL values.
- Each table can have only one primary key.
- It ensures data integrity and prevents duplicate records.

Code:

Case 1: Valid Insert

```
Ananya_DB
Worksheet   Query Builder

CREATE TABLE Department (
    Dept_ID NUMBER PRIMARY KEY,
    Dept_Name VARCHAR2(50)
);
INSERT INTO Department VALUES (1, 'CSE');
```

Output:

```
Table DEPARTMENT created.


1 row inserted.
```

Code:

Case 2: Constraint Violated

```
Ananya_DB
Worksheet   Query Builder

INSERT INTO Department VALUES (1, 'IT');
```

Output:

```
INSERT INTO Department VALUES (1, 'IT')
Error report -
ORA-00001: unique constraint (SYSTEM.SYS_C008551) violated
```

Practical-05

Implementation of Business Constraint: Null, Not Null, Default, Check.

Objective: - To implement business constraints such as NULL, NOT NULL, DEFAULT, and CHECK to enforce rules and maintain data validity in a database.

NULL Constraint: -

- The NULL constraint allows a column to store empty or missing values.
- It is used when a value is optional and not mandatory.

Code:

```
CREATE TABLE Emp (
    Emp_ID NUMBER PRIMARY KEY,
    Name VARCHAR2(50) NOT NULL,
    Age NUMBER CHECK (Age >= 18),
    Salary NUMBER DEFAULT 10000,
    Email VARCHAR2(50)
);
INSERT INTO Emp VALUES (1, 'Riya', 22, 25000, NULL);
```

Output:

```
Table EMP created.


1 row inserted.
```

NOT NULL Constraint: -

- The NOT NULLconstraint ensures that a column must always contain a value.
- It prevents insertion of empty or missing data in that column.